

A Validated Machine Learning Surrogate for Rapid Forward and Reverse Ion Implantation Modeling in Gallium Arsenide Substrates for Plasma Based Semiconductor Industry

Abdullah Shadek Fahim¹, fahim.just.19@gmail.com, Abu Taher¹, abutaher.eee.just@gmail.com, Musavvir Haque¹, musavvirhaquepial@gmail.com, Md. Rashed Khan, rashedkhan039@gmail.com, Mahabub Hasan Rocky¹, rocky111461@gmail.com, Md. Amzad Hossain¹, ma.hossain@just.edu.bd.

¹ *Interdisciplinary Laboratory of Advanced Materials Processing and Simulation (I LAMPS), Department of Electrical and Electronic Engineering, Jashore University of Science and Technology, Jashore - 7408, Bangladesh*

Gallium Arsenide (GaAs) is still an important material for high-speed RF and optoelectronic devices, however ion implantation profiles in Binary compound GaAs are challenging to predict. For heavy substrates, traditional Monte Carlo simulations like Stopping and Range of Ions in Matter (SRIM) are too computationally expensive because of extremely dense nuclear collision cascades while analytical models (eg. Pearson-IV) often do not account for complex boundary effects like backscattering and stoichiometric mismatch. To facilitate data-driven and rapid prediction of implantation dynamics, we introduce a Dual-Stage Random Forest framework for the first time in this work which is developed as a fast high-fidelity surrogate model specifically dedicated to predict GaAs. The model was trained on a robust dataset comprised of 5,600 simulations for five ion species (Boron, Phosphorus, Arsenic, Magnesium and Argon) considering the energy range between 10 keV to 10 MeV. The approach uses a classifier to recognize transmission events and a multi-output regressor to estimate nine relevant physical parameters (i.e., Projected Range, Straggling and Vacancy production). The results show that the surrogate obtains a global R-squared score just above 0.94 for heavy ions (the case of As in GaAs), capturing adequately the deterministic nuclear stopping regime. The model results show good agreement (99% fidelity) in the prediction of backscattering, which is an important parameter for dose control in processing of thick substrate. Moreover, a computational complexity analysis shows that 20000 times speed-up over conventional Monte Carlo simulations (0.015 s versus 385 s for simulating 10,000 ions), which allows for real-time inverse design. This work charts a path for Digital Twin inclusion towards GaAs process optimization, overcoming the gap between physical realism and computational throughput in plasma-based semiconductor production.